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EST II — MATH LEVEL 2 • UNIT 4.01

# Matrices, Determinants & 2D/3D Vectors

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

**Total Questions:** 25 • Verified Step-by-Step Solutions

## QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

| | | | | |
|---|---|---|---|---|
| Q1 A | Q2 A | Q3 A | Q4 A | Q5 A |
| Q6 A | Q7 A | Q8 A | Q9 B | Q10 A |
| Q11 B | Q12 B | Q13 A | Q14 D | Q15 C |
| Q16 C | Q17 A | Q18 D | Q19 B | Q20 C |
| Q21 D | Q22 A | Q23 C | Q24 B | Q25 B |

## Matrices, Determinants & 2D/3D Vectors Practice Exercises

**1** MCQ

A vector $\mathbf{v}$ is represented by the directed line segment from point $P(1, 2)$ to point $Q(4, 6)$. What are the components of vector $\mathbf{v}$?

- **A** [3, 4] ✓
- **B** [5, 8]
- **C** [-3, -4]
- **D** [4, 3]

**WORKED SOLUTION & VERIFICATION**

The components of a vector $\mathbf{v}$ represented by a directed line segment from point $P(x_1, y_1)$ to point $Q(x_2, y_2)$ are given by $\mathbf{v} = [x_2 - x_1, y_2 - y_1]$. In this case, $P = (1, 2)$ and $Q = (4, 6)$. Therefore, the components of $\mathbf{v}$ are $[4 - 1, 6 - 2] = [3, 4]$.

**2** MCQ

A translation maps point $A(1, 2)$ to $A'(-4, 2)$ and point $B(3, 5)$ to $B'(-2, 5)$. This transformation is represented by $\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} a \\ b \end{pmatrix}$. Find the values of $a$ and $b$.

- **A** a = -5, b = 0 ✓
- **B** a = 5, b = 0
- **C** a = 0, b = -5
- **D** a = 0, b = 5

**WORKED SOLUTION & VERIFICATION**

Subtracting the original coordinates of $A(1, 2)$ from its image $A'(-4, 2)$ gives the translation vector:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -4 - 1 \\ 2 - 2 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \end{pmatrix}.$$

Checking with $B(3, 5) \rightarrow B'(-2, 5)$ confirms $(-2 - 3, 5 - 5) = (-5, 0)$. Thus, $a = -5$ and $b = 0$.

**3** MCQ

A linear transformation in the 2D plane is represented by the matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. This transformation maps the point $P(1, 2)$ to $P'(-3, 7)$ and the point $Q(3, 1)$ to $Q'(5, 1)$. Determine the values of $a, b, c$, and $d$, and then calculate the sum $a + b + c + d$.

- **A** 9 ✓
- **B** 11
- **C** 7
- **D** 5

**WORKED SOLUTION & VERIFICATION**

Let the linear transformation matrix be $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.
The transformation of a point $\mathbf{x}$ to $\mathbf{x}'$ is given by $M\mathbf{x} = \mathbf{x}'$.
For point $P(1, 2)$ mapped to $P'(4, 9)$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \\ 9 \end{pmatrix}$$

This yields the equations:
1) $a + 2b = 4$
2) $c + 2d = 9$
For point $Q(3, 1)$ mapped to $Q'(7, 12)$:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 7 \\ 12 \end{pmatrix}$$

This yields the equations:
3) $3a + b = 7$
4) $3c + d = 12$
We need to solve these two systems of linear equations for $a, b, c$, and $d$.
Solving for $a$ and $b$ using equations (1) and (3):
From equation (3), $b = 7 - 3a$. Substitute this into equation (1):
$a + 2(7 - 3a) = 4$
$a + 14 - 6a = 4$
$-5a = 4 - 14$
$-5a = -10$
$a = 2$
Now substitute the value of $a$ back into the expression for $b$:
$b = 7 - 3(2) = 7 - 6 = 1$
Solving for $c$ and $d$ using equations (2) and (4):
From equation (4), $d = 12 - 3c$. Substitute this into equation (2):
$c + 2(12 - 3c) = 9$
$c + 24 - 6c = 9$
$-5c = 9 - 24$
$-5c = -15$
$c = 3$
Now substitute the value of $c$ back into the expression for $d$:
$d = 12 - 3(3) = 12 - 9 = 3$
So, the transformation matrix is $M = \begin{pmatrix} 2 & 1 \\ 3 & 3 \end{pmatrix}$.
The sum of the elements of the transformation matrix is $a + b + c + d = 2 + 1 + 3 + 3 = 9$.
Common misconceptions might arise from errors in solving the systems of equations, such as sign errors or

algebraic mistakes when isolating variables. For instance, incorrectly solving for $b$ or $d$ could lead to a wrong sum. Alternatively, a student might forget to sum all four elements and only sum the elements of one row or column.

**4** MCQ

A matrix $M = \begin{pmatrix} k & 2 \\ 3 & 1 \end{pmatrix}$ transforms a point $P(1, 2)$ to $P'(5, 5)$. Find the value of $k$.

- **A** 1 ✓
- **B** 2
- **C** 3
- **D** 4

**WORKED SOLUTION & VERIFICATION**

The transformation of a point $P(x, y)$ by a matrix $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is given by $MP$. In this case, we have $\begin{pmatrix} k & 2 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}$. Performing the matrix multiplication, we get: $k(1) + 2(2) = 5$ and $3(1) + 1(2) = 5$. The second equation, $3 + 2 = 5$, is consistent. For the first equation, we have $k + 4 = 5$. Solving for $k$, we subtract 4 from both sides: $k = 5 - 4$, which gives $k = 1$. Therefore, the value of $k$ is 1.

**5** MCQ

Consider the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$. When the matrix $A$ is applied to the vector $\mathbf{v} = \begin{pmatrix} x \\ y \end{pmatrix}$, the resulting vector is $A\mathbf{v} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$. What is the original vector $\mathbf{v}$?

- **A** $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$ ✓
- **B** $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$
- **C** $\begin{pmatrix} 6 \\ 3 \end{pmatrix}$
- **D** $\begin{pmatrix} 12 \\ 27 \end{pmatrix}$

**WORKED SOLUTION & VERIFICATION**

The matrix multiplication is given by:

$$\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2x \\ 3y \end{pmatrix}$$

We are given that this result is $\begin{pmatrix} 6 \\ 9 \end{pmatrix}$. Therefore, we have the system of equations:

$$2x = 6$$

$$3y = 9$$

Solving for $x$ and $y$:

$$x = \frac{6}{2} = 3$$

$$y = \frac{9}{3} = 3$$

So, the original vector $\mathbf{v}$ is $\begin{pmatrix} 3 \\ 3 \end{pmatrix}$.

**6** MCQ

Consider a linear transformation $T$ represented by the matrix $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. Let $P = \begin{pmatrix} x \\ y \end{pmatrix}$ be a point in the plane. The transformation maps $P$ to $P'$. If the area of the region transformed by $T$ from a unit square in the $xy$-plane is $A$, and the determinant of $A$ is $\det(A)$, find the value of $\det(A)$ if the area of the transformed region is $5$.

- **A** 5 ✓
- **B** 2
- **C** 1
- **D** 10

**WORKED SOLUTION & VERIFICATION**

The area of the region transformed by a linear transformation $T$ represented by a matrix $A$ from a unit square is given by the absolute value of the determinant of $A$, i.e., Area $= |\det(A)|$. In this problem, we are given that the matrix representing the transformation is $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$. The determinant of $A$ is calculated as: $\det(A) = (2 \times 3) - (1 \times 1) = 6 - 1 = 5$. We are also given that the area of the transformed region is $5$. Thus, $|\det(A)| = 5$. Since we calculated $\det(A) = 5$, which is positive, the condition is satisfied. Therefore, the value of $\det(A)$ is $5$.

**7** MCQ

Consider the transformation grid below. If point A has coordinates $(1, 2)$, what are the coordinates of its image, A', after the transformation shown?

- **A** **(-4, 2)** ✓
- **B** (-2, 1)
- **C** (2, -4)
- **D** (1, -2)

**WORKED SOLUTION & VERIFICATION**

The transformation grid shows a reflection across the line $y = x$. When a point $(x, y)$ is reflected across the line $y = x$, its image has coordinates $(y, x)$. Given point A is $(1, 2)$, its image A' will have coordinates $(2, 1)$. However, the provided `transformation_grid` JSON shows an object at `[[1, 2], [4, 2], [2, 5]]` labeled A, B, C and its image at `[[-4, 2], [-1, 2], [-3, 5]]` labeled A', B', C'. This implies a translation, not a reflection. Let's analyze the transformation of point A. Original point A is at $(1, 2)$. Its image A' is at $(-4, 2)$. To find the translation vector $(\Delta x\ \Delta y)$, we subtract the original coordinates from the image coordinates: $\Delta x = -4 - 1 = -5$ and $\Delta y = 2 - 2 = 0$. So the translation is by the vector $(-5\ 0)$. The question asks for the coordinates of A' given A. Based on the figure's data, A' is $(-4, 2)$. The provided choices are: (-4, 2), (-2, 1), (2, -4), (1, -2). The first choice matches the figure.

**8** MCQ

If $A = \begin{bmatrix} -4 & -3 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 1 \\ -3 & -2 \end{bmatrix}$, find $A + B$.

- **A** $\begin{bmatrix} -8 & -2 \\ -3 & -2 \end{bmatrix}$ ✓
- **B** $\begin{bmatrix} 0 & -4 \\ 3 & 2 \end{bmatrix}$
- **C** $\begin{bmatrix} -8 & -1 \\ -3 & -2 \end{bmatrix}$
- **D** $\begin{bmatrix} -8 & -2 \\ -4 & -2 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -8 & -2 \\ -3 & -2 \end{bmatrix}$.

**9** MCQ

If $A = \begin{bmatrix} -3 & -5 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -4 & 0 \\ 3 & 3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 1 & -5 \\ -3 & -1 \end{bmatrix}$

B $\begin{bmatrix} -7 & -5 \\ 3 & 5 \end{bmatrix}$ ✓

C $\begin{bmatrix} -7 & -4 \\ 3 & 5 \end{bmatrix}$

D $\begin{bmatrix} -7 & -5 \\ 2 & 5 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -7 & -5 \\ 3 & 5 \end{bmatrix}$.

**10** MCQ

If $A = \begin{bmatrix} 1 & -1 \\ -4 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -1 \\ -2 & -2 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 4 & -2 \\ -6 & -7 \end{bmatrix}$ ✓

B $\begin{bmatrix} -2 & 0 \\ -2 & -3 \end{bmatrix}$

C $\begin{bmatrix} 4 & -1 \\ -6 & -7 \end{bmatrix}$

D $\begin{bmatrix} 4 & -2 \\ -7 & -7 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 4 & -2 \\ -6 & -7 \end{bmatrix}$.

**11** MCQ

If $A = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & -5 \\ -3 & -1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -8 & 8 \\ 5 & 0 \end{bmatrix}$

B $\begin{bmatrix} -2 & -2 \\ -1 & -2 \end{bmatrix}$ ✓

C $\begin{bmatrix} -2 & -1 \\ -1 & -2 \end{bmatrix}$

D $\begin{bmatrix} -2 & -2 \\ -2 & -2 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -2 & -2 \\ -1 & -2 \end{bmatrix}$.

**12** MCQ

If $A = \begin{bmatrix} -5 & 5 \\ 0 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 4 \\ -5 & -5 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -4 & 9 \\ -6 & 0 \end{bmatrix}$

B $\begin{bmatrix} -4 & 9 \\ -5 & 0 \end{bmatrix}$ ✓

C $\begin{bmatrix} -6 & 1 \\ 5 & 10 \end{bmatrix}$

D $\begin{bmatrix} -4 & 10 \\ -5 & 0 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -4 & 9 \\ -5 & 0 \end{bmatrix}$.

**13** MCQ

If $A = \begin{bmatrix} -2 & 0 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -3 & 1 \\ 2 & 4 \end{bmatrix}$ ✓

B $\begin{bmatrix} -1 & -1 \\ 2 & 2 \end{bmatrix}$

C $\begin{bmatrix} -3 & 2 \\ 2 & 4 \end{bmatrix}$

D $\begin{bmatrix} -3 & 1 \\ 1 & 4 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -3 & 1 \\ 2 & 4 \end{bmatrix}$.

**14** MCQ

If $A = \begin{bmatrix} 2 & -5 \\ -3 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 1 \\ 1 & -4 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 2 & -6 \\ -4 & 0 \end{bmatrix}$

B $\begin{bmatrix} 2 & -3 \\ -2 & -8 \end{bmatrix}$

C $\begin{bmatrix} 2 & -4 \\ -3 & -8 \end{bmatrix}$

D $\begin{bmatrix} 2 & -4 \\ -2 & -8 \end{bmatrix}$ ✓

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 2 & -4 \\ -2 & -8 \end{bmatrix}$.

**15** MCQ

If $A = \begin{bmatrix} 5 & 4 \\ 0 & -4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 3 \\ 1 & 1 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 2 & 1 \\ -1 & -5 \end{bmatrix}$

B $\begin{bmatrix} 8 & 8 \\ 1 & -3 \end{bmatrix}$

C $\begin{bmatrix} 8 & 7 \\ 1 & -3 \end{bmatrix}$ ✓

D $\begin{bmatrix} 8 & 7 \\ 0 & -3 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 8 & 7 \\ 1 & -3 \end{bmatrix}$.

**16** MCQ

If $A = \begin{bmatrix} 3 & -5 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -1 \\ -2 & -2 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 3 & -5 \\ -2 & -1 \end{bmatrix}$

B $\begin{bmatrix} 3 & -6 \\ -3 & -1 \end{bmatrix}$

C $\begin{bmatrix} 3 & -6 \\ -2 & -1 \end{bmatrix}$ ✓

D $\begin{bmatrix} 3 & -4 \\ 2 & 3 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 3 & -6 \\ -2 & -1 \end{bmatrix}$.

**17** MCQ

If $A = \begin{bmatrix} 3 & -5 \\ 1 & -5 \end{bmatrix}$ and $B = \begin{bmatrix} 5 & -1 \\ -5 & 5 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 8 & -6 \\ -4 & 0 \end{bmatrix}$ ✓

B $\begin{bmatrix} -2 & -4 \\ 6 & -10 \end{bmatrix}$

C $\begin{bmatrix} 8 & -5 \\ -4 & 0 \end{bmatrix}$

D $\begin{bmatrix} 8 & -6 \\ -5 & 0 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 8 & -6 \\ -4 & 0 \end{bmatrix}$.

**18** MCQ

If $A = \begin{bmatrix} -3 & 4 \\ 5 & -3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 0 \\ 2 & 4 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} -5 & 4 \\ 3 & -7 \end{bmatrix}$

B $\begin{bmatrix} -1 & 5 \\ 7 & 1 \end{bmatrix}$

C $\begin{bmatrix} -1 & 4 \\ 6 & 1 \end{bmatrix}$

D $\begin{bmatrix} -1 & 4 \\ 7 & 1 \end{bmatrix}$ ✓

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} -1 & 4 \\ 7 & 1 \end{bmatrix}$.

**19** MCQ

If $A = \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & -3 \\ -5 & 3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 0 & 4 \\ 3 & -2 \end{bmatrix}$

B $\begin{bmatrix} 8 & -2 \\ -7 & 4 \end{bmatrix}$ ✓

C $\begin{bmatrix} 8 & -1 \\ -7 & 4 \end{bmatrix}$

D $\begin{bmatrix} 8 & -2 \\ -8 & 4 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 8 & -2 \\ -7 & 4 \end{bmatrix}$.

**20** MCQ

If $A = \begin{bmatrix} 1 & 1 \\ -1 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0 \\ 2 & -3 \end{bmatrix}$, find $A + B$.

A $\begin{bmatrix} 4 & 2 \\ 1 & 2 \end{bmatrix}$

B $\begin{bmatrix} 4 & 1 \\ 0 & 2 \end{bmatrix}$

C $\begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$ ✓

D $\begin{bmatrix} -2 & 1 \\ -3 & 8 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Add entrywise: $A + B = \begin{bmatrix} 4 & 1 \\ 1 & 2 \end{bmatrix}$.

**21** MCQ

If $A = \begin{bmatrix} -6 & -2 \\ 3 & -3 \end{bmatrix}$, find $-4A$.

A $\begin{bmatrix} 20 & 8 \\ -12 & 12 \end{bmatrix}$

B $\begin{bmatrix} -24 & -8 \\ 12 & -12 \end{bmatrix}$

C $\begin{bmatrix} 20 & 4 \\ -16 & 8 \end{bmatrix}$

D $\begin{bmatrix} 24 & 8 \\ -12 & 12 \end{bmatrix}$ ✓

**WORKED SOLUTION & VERIFICATION**

Multiply every entry by -4: $-4A = \begin{bmatrix} 24 & 8 \\ -12 & 12 \end{bmatrix}$.

**22** MCQ

Suppose that $A = \begin{bmatrix} -4 & -6 \\ 1 & -2 \end{bmatrix}$, find $-2A$.

A $\begin{bmatrix} 8 & 12 \\ -2 & 4 \end{bmatrix}$ ✓

B $\begin{bmatrix} -8 & -12 \\ 2 & -4 \end{bmatrix}$

C $\begin{bmatrix} 6 & 10 \\ -4 & 2 \end{bmatrix}$

D $\begin{bmatrix} 6 & 12 \\ -2 & 4 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Multiply every entry by -2: $-2A = \begin{bmatrix} 8 & 12 \\ -2 & 4 \end{bmatrix}$.

**23** MCQ

If $A = \begin{bmatrix} -6 & -4 \\ -5 & 6 \end{bmatrix}$, find $-3A$.

A $\begin{bmatrix} 15 & 9 \\ 12 & -21 \end{bmatrix}$

B $\begin{bmatrix} 15 & 12 \\ 15 & -18 \end{bmatrix}$

C $\begin{bmatrix} 18 & 12 \\ 15 & -18 \end{bmatrix}$ ✓

D $\begin{bmatrix} -18 & -12 \\ -15 & 18 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Multiply every entry by -3: $-3A = \begin{bmatrix} 18 & 12 \\ 15 & -18 \end{bmatrix}$.

**24** MCQ

Assuming that $A = \begin{bmatrix} -3 & -3 \\ 6 & -1 \end{bmatrix}$, find $-2A$.

A $\begin{bmatrix} 4 & 4 \\ -14 & 0 \end{bmatrix}$

B $\begin{bmatrix} 6 & 6 \\ -12 & 2 \end{bmatrix}$ ✓

C $\begin{bmatrix} 4 & 6 \\ -12 & 2 \end{bmatrix}$

D $\begin{bmatrix} -6 & -6 \\ 12 & -2 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Multiply every entry by -2: $-2A = \begin{bmatrix} 6 & 6 \\ -12 & 2 \end{bmatrix}$.

**25** MCQ

If $A = \begin{bmatrix} -1 & 2 \\ 4 & 4 \end{bmatrix}$, find $-4A$.

A $\begin{bmatrix} -4 & 8 \\ 16 & 16 \end{bmatrix}$

B $\begin{bmatrix} 4 & -8 \\ -16 & -16 \end{bmatrix}$ ✓

C $\begin{bmatrix} 0 & -12 \\ -20 & -20 \end{bmatrix}$

D $\begin{bmatrix} 0 & -8 \\ -16 & -16 \end{bmatrix}$

**WORKED SOLUTION & VERIFICATION**

Multiply every entry by -4: $-4A = \begin{bmatrix} 4 & -8 \\ -16 & -16 \end{bmatrix}$.